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Evaluating the Effect of Play-Action Passing on Defensive Disruption

My main question is whether play-action is associated with a different defensive reaction at the snap, and whether that difference shows up in pressure and in offensive efficiency via EPA. Offenses have leaned more and more on play-action in the past few years, and while it is widely accepted that it works (Sharp, 2019), there is still a pretty large gap in understanding exactly why. My project looks to really evaluate the effect that play-action has on a defense through measuring what the defense does at the snap and whether that change shows up in pressure and in offensive efficiency.

The data comes from the NFL Big Data Bowl on Kaggle (National Football League, 2024) and covers weeks 1 through 9 of the 2022 season, from the 2025 competition. I kept plays on downs one to three with ten or fewer yards to go, and removed quarterback spikes, plays nullified by a penalty, overtime, and garbage time. I defined garbage time as the fourth quarter with a score margin above sixteen points in either direction. This left me with 7,368 plays, made up of 5,243 standard dropbacks and 2,125 play-action dropbacks.

The main explanatory variable I used is a true-or-false indicator for whether the offense used play-action. To measure how the defense reacted, I used pass rusher get-off time, which measures the number of seconds a pass rusher takes to cross the line of scrimmage after the snap.

I also wanted to confirm that a slower start would translate into less pressure, so I used the data's pressure flag for each rusher and marked a play as pressured if any of the rushers were able to cause a pressure. Finally, I measured offensive efficiency with EPA.

I first ran a multiple linear regression on all of the plays:

$$Y = \beta0 + \beta1(\text{PlayAction}) + \beta2(\text{Down2}) + \beta3(\text{Down3}) + \beta4(\text{YardsToGo}) + \beta5(\text{AbsoluteYardline}) + \beta6(\text{ScoreDiff}) + \varepsilon$$

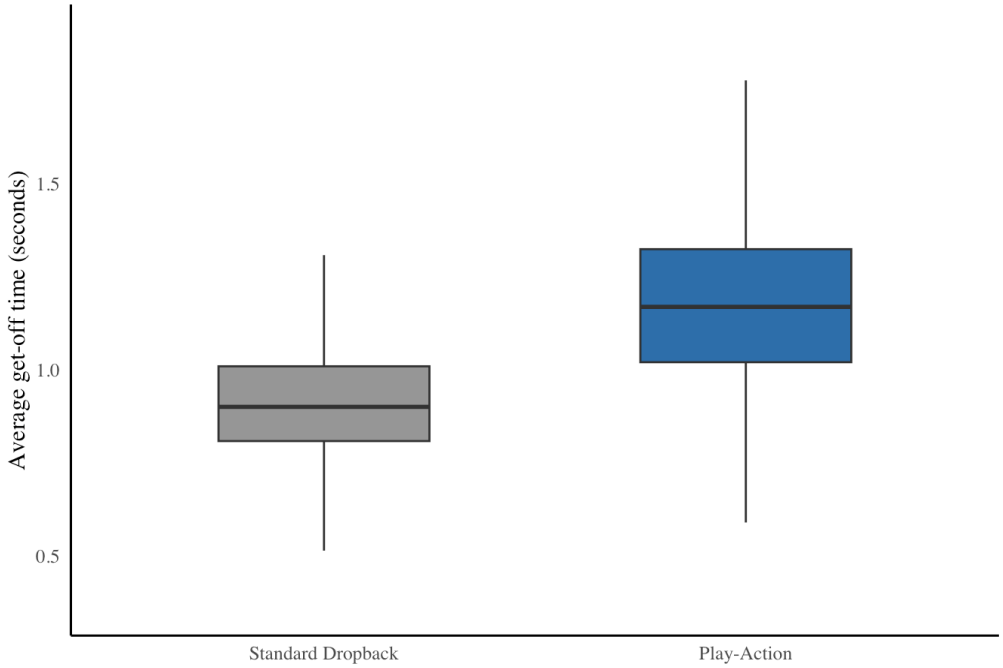
where Y is get-off time or EPA, and β1 is the difference between a play-action play and a standard dropback after accounting for the other variables. Down is entered as two binary variables, meaning Down2 and Down3 compare second and third down to first. Because pressure is binary, I did a logistic regression and used the same right side of the equation. I also matched plays on down, distance, field position, and score.

I ran the multiple linear regression once for both outcomes. For get-off time, β1 was 0.226 seconds, with a 95% confidence interval from 0.217 to 0.236, so rushers got off the line about a quarter of a second slower on play-action. Which is a good first sign that play-action changes what the defense does at the snap. For EPA, β1 was 0.087 per dropback, with a 95% confidence interval from 0.003 to 0.171. So while an offense does gain a small amount of expected points each time it uses play-action, the interval is really close to zero, making the result less certain. Compared to first down, rushers got off about 0.04 seconds faster on second down, where β2 was −0.039, and about 0.12 seconds faster on third down, where β3 was −0.115. This makes sense since defenses are usually expecting a pass on later downs. Third-down dropbacks also had lower EPA than first-down dropbacks, with a β3 of −0.17, which also makes sense because it is a lot harder to be efficient when the defense knows what is coming. Field position did not make

much of an impact on either outcome. The get-off model did explain a good amount of the variation, with an R^2 of 0.31, meaning play type and situation account for about a third of why get-off times differ. However, the EPA model explains almost none, with an R^2 of just 0.003, meaning EPA is mostly just noise. This shows that the parts of football the model cannot interpret, like player talent, personnel, and coverage, still end up deciding most plays' EPA.

Figure 1 helps show that this pass rush get-off slowdown is not just driven by a few unusual plays. The typical play-action get-off was 1.17 seconds compared to 0.90 seconds on standard dropbacks, and the entire box of play-action plays sits completely above the box of standard dropbacks. The slowdown also showed up on every down and for all 32 of the teams' defenses, meaning it is seen league-wide, which adds to the results' consistency.

Figure 1. Pass Rushers Get Off the Line Slower on Play-Action

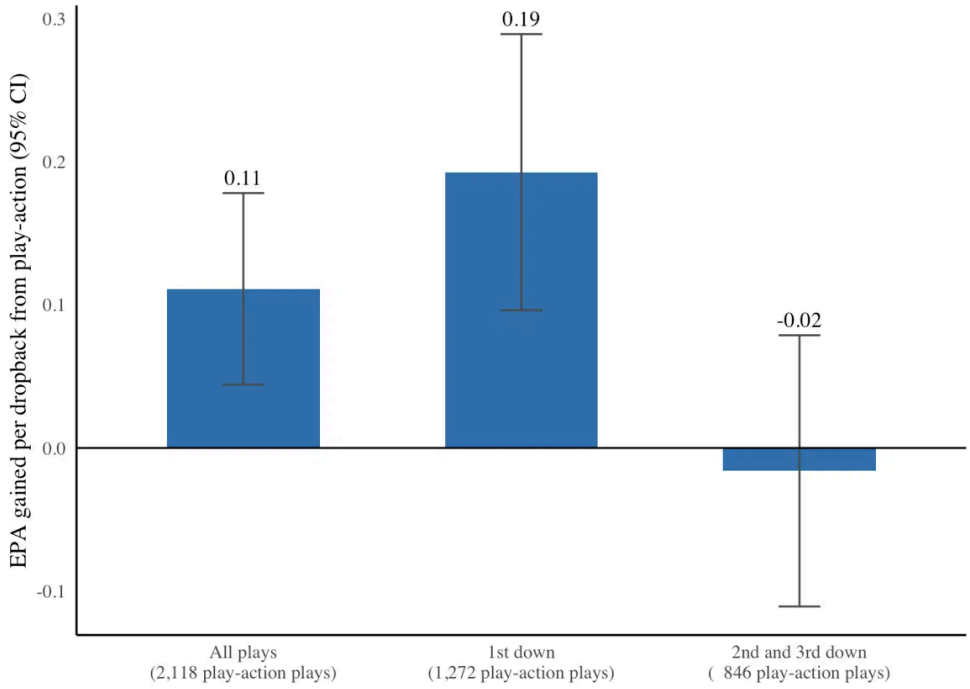


Average get-off time by the play type (standard n = 5,217; play-action n = 2,100; 51 plays missing get-off excluded). Boxes show the middle 50% of plays and lines show medians; outliers hidden. Weeks 1 to 9 of the 2022 NFL season.

Slower get-offs went with less pressure, and pressure is costly for offenses, as a pressured dropback loses about 0.53 EPA, with a 95% confidence interval from −0.61 to −0.46. However, play-action was also associated with a higher rate of pressure, with an odds ratio of 1.22 and a 95% confidence interval from 1.09 to 1.37. One possible reason could be that most play-action dropbacks take longer to develop, giving the rushers time to recover, something that would be worth exploring but is hard to test.

The next question is whether play-action actually adds expected points. Offenses run the ball on first downs at around a 50% rate, which makes first down basically a coin flip between run and pass (Wilk, 2026). This means first down is the down where a defense is most likely to expect the run, and where a fake handoff should therefore be at its most effective. The results mirror this, as EPA gain on play-action dropbacks was 0.193 on first down, with a 95% confidence interval from 0.096 to 0.289, and −0.016 on second and third down combined, with a 95% confidence interval from −0.111 to 0.079 (Figure 2). The first-down gain is much larger, and its interval stays well above zero, while the late-down intervals contain zero, meaning there is no statistically significant gain. With a p-value of .003, this gap is statistically significant, showing that the overall gain is really coming from mostly first down and not from each down equally. I also wanted to see if third-and-short could be just as effective, since a run is a realistic option there, similar to first down. Unfortunately, there were only 83 plays that fit that situation, and the estimated gain of 0.07 EPA had an interval from −0.35 to 0.48, which is far too wide to say anything reliable, most likely due to the extremely small sample size.

Figure 2. Play-Action Adds EPA on First Down but Shows No Clear Gain on Late Downs



Estimated EPA gained per dropback from play-action among matched plays, with 95% confidence intervals. Labels show the number of play-action plays. Second and third down are combined. Weeks 1 to 9 of the 2022 NFL season.

There are some clear limitations I have faced so far with this project. For starters, these results are from just nine weeks of one season. Plays within a game are not independent, and unusual plays such as trick plays and run-pass options were not filtered as there is no way to differentiate them, meaning some of the plays in the data may not behave like normal dropbacks. Because play calling is not random, I am not able to claim true cause and effect, so every result in this paper will be an association, not a causal effect. A slower get-off could also come from things other than defensive hesitation, such as blockers firing out the way they would on a run, so though I can say play-action goes with a slower rush, I cannot say for sure why. My splits by down and by pressure were chosen after I saw the first results, so they should be treated as

exploratory. Another limitation is the sample size in specific situations like third and short, which only had 83 play-action plays. That sample size is very small and created a confidence interval that was far too wide to draw any reliable conclusions, leaving that question unanswered. I also used an AI tool, Claude (Anthropic, 2026), to help write and debug the R code. I ran the code myself and checked the results against the output, but AI-written code can contain mistakes or choices I did not fully think through, so my results are only as reliable as that code. Moving forward, I plan to attempt to work with the player tracking data to measure defender movement directly. This should add even more context behind my analysis. I also plan to look at the third and short idea again, by looking at short yardage on any down or by adding more seasons of data, which would give me more plays without loosening the controls.

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